FRIDAY 6 NOVEMBER 2026 · 09:00–10:00 · GROUP 07

Linear Algebra

Linear maps, their three forms, kernel, image and rank–nullity · FRY BLDG G.14

Try a first step before each board discussion. Write definitions and reasons, not just final answers. Question labels match the board.

START HERE · 01–04 MINUTES

Key ideas: linear maps, kernel, image, rank

Linear Algebra · Week 7 · Chapter 7 · Keep this page beside your working.

1 · What "linear" means

Let V, W be vector spaces over a field K. A function $T : V \to W$ is **K-linear** (Definition 7.1) if for all $v, w \in V$ and $\lambda \in K$:

$$T(v + w) = T(v) + T(w), \qquad T(\lambda v) = \lambda T(v).$$

To prove linear: check both identities for *arbitrary* vectors and scalars. **To prove not linear:** one explicit counterexample is enough. A quick first test: every linear map has $T(0) = 0$ (Lemma 7.4(i)) — but $T(0) = 0$ alone does *not* prove linearity (e.g. $(x, y) \mapsto (0, \sin x)$).

The field matters. A map $\mathbb{C}^n \to \mathbb{C}^m$ can be $\mathbb{R}$-linear without being $\mathbb{C}$-linear: test the scalar $\lambda = i$.

2 · Three forms of the same linear map

Form	Example: $f(x, y, z) = (3x,\ x + y + z)$
Algebraic form	$f(x, y, z) = (3x,\ x + y + z)$
Standard basis form	$f(e_1) = 3e_1 + e_2,\ f(e_2) = e_2,\ f(e_3) = e_2$
Matrix form	$f(x) = Ax,\ A = \begin{pmatrix} 3 & 0 & 0 \\ 1 & 1 & 1 \end{pmatrix} \in M_{2,3}(\mathbb{R})$

The Fact (7.14): Ae_i is the ith column of A. So the columns of the matrix are $f(e_1), \ldots, f(e_n)$, and a linear map is determined by where it sends $e_1, \ldots, e_n$ (Remark 7.10). Size check: $f : \mathbb{R}^n \to \mathbb{R}^m$ has an $m \times n$ matrix. Composition matches multiplication: the matrix of $h \circ g$ is $A_h A_g$ (apply g first).

Warning (Example 7.20). You can always build a matrix from $f(e_1), \ldots, f(e_n)$; it represents f only if it agrees with f on *every* input.

3 · Kernel, image, rank and nullity

$$\ker T = \{v \in V : T(v) = 0\}, \qquad \operatorname{Im} T = \{T(v) : v \in V\}, \qquad \operatorname{nullity} T = \dim \ker T, \quad \operatorname{rank} T = \dim \operatorname{Im} T.$$

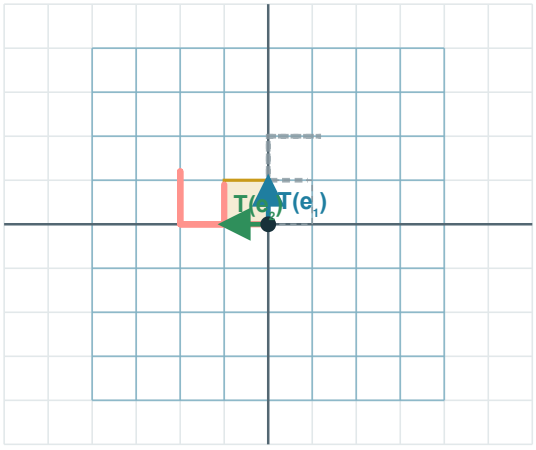
Both are subspaces (Lemma 7.27). T is injective $\iff \ker T = \{0\}$; T is surjective $\iff \operatorname{Im} T = W$. For $T(x) = Ax$, $\operatorname{Im} T$ is the span of the columns and $\ker T = S(A, 0)$.

Rank–Nullity Theorem (Theorem 7.31). For $T : V \to W$ with $\dim V = n$ finite, $\operatorname{rank} T + \operatorname{nullity} T = n = \dim V$. Count dimensions over the stated field: $\dim_{\mathbb{C}} \mathbb{C}^4 = 4$ but $\dim_{\mathbb{R}} \mathbb{C}^4 = 8$.

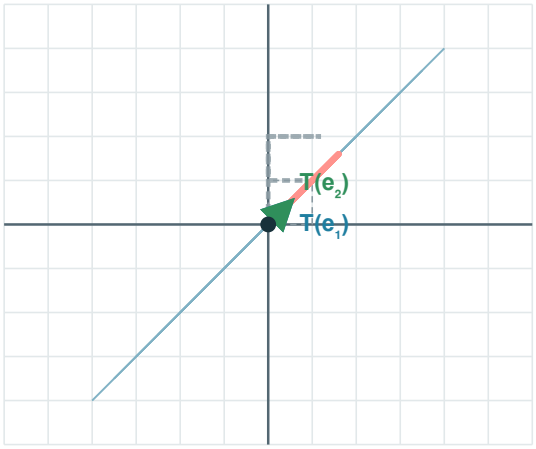
VISUAL · USE DURING L1, L3–L4 AND L6

What a linear map does to the grid

Dashed: the unit square and the letter F before the map. Solid: their images. The blue and green arrows are $T(e_1)$, $T(e_2)$ — the columns of the matrix (Fact 7.14).



Matrix columns = images of e_1, e_2: (0, 1) and (-1, 0)



Matrix columns = images of e_1, e_2: (0.5, 0.5) and (0.5, 0.5)

Q8 g: rotation π/2: Columns are $g(e_1) = e_2$ and $g(e_2) = -e_1$.

Tutorial questions

L1 · Definition of a linear map: which of these four are linear?

LA tutor guide Week 7 (R. Carey, 29 Oct 2026): Definition of a linear map; Definition 7.1, Lemma 7.4(i), Example 7.22

(a) State the definition of an $\mathbb{R}$-linear map $T : \mathbb{R}^2 \to \mathbb{R}^2$. How will you remember it?

(b) Decide whether each of the following maps $\mathbb{R}^2 \to \mathbb{R}^2$ is $\mathbb{R}$-linear:

$$T_1 : \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x+2 \\ y+2 \end{pmatrix}, \quad T_2 : \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} 2x \\ 3y \end{pmatrix}, \quad T_3 : \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x+y \\ x-y \end{pmatrix}, \quad T_4 : \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x+y \\ y^2 \end{pmatrix}.$$

(c) What do the linear ones have in common, and what goes wrong in the others? Suggest a general form for a linear map $\mathbb{R}^2 \to \mathbb{R}^2$.

(d) Lemma 7.4(i) says that a linear map satisfies $T(0) = 0$. Show that $g : \mathbb{R}^2 \to \mathbb{R}^2$, $g(x, y) = (0, \sin x)$, satisfies $g(0) = 0$ but is not linear.

L2 · Challenge: one condition holds, the other fails

LA tutor guide Week 7 (R. Carey, 29 Oct 2026): challenge in footnote 8; links to PS6 Q3(iii)

A map can satisfy one condition of Definition 7.1 but not the other.

(a) Let $f : \mathbb{C} \to \mathbb{C}$, $f(z) = \overline{z}$ (complex conjugation). Show that $f(z + w) = f(z) + f(w)$ for all $z, w \in \mathbb{C}$, but that $f(\lambda z) = \lambda f(z)$ fails for some $\lambda \in \mathbb{C}$. Is f $\mathbb{R}$-linear?

(b) Let $f : \mathbb{R}^2 \to \mathbb{R}$ be given by

$$f(x, y) = \begin{cases} x & \text{if } y = 0, \\ y & \text{if } y \neq 0. \end{cases}$$

Show that $f(\lambda v) = \lambda f(v)$ for all $\lambda \in \mathbb{R}$ and $v \in \mathbb{R}^2$, but that f is not additive.

L3 · Three forms of a linear map: Q7 and Q8 (rotation by π/2 and its inverse)

LA PS6 Q7(a)–(c), Q8(a)–(f); pair activity from LA tutor guide Week 7 (R. Carey, 29 Oct 2026)

Q7. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $f(x, y) = (-x, -y)$ for all $x, y \in \mathbb{R}$.
(a) Write f in standard basis form.
(b) Write f in matrix form.
(c) Using the matrix form, check that f is indeed an $\mathbb{R}$-linear map.

Q8. Let $g : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear map with $g(e_1) = e_2$ and $g(e_2) = -e_1$.
(a) Does the information above enable us to find, uniquely, $g(x)$ for all $x \in \mathbb{R}^2$? Justify your answer.
(b) Write g in algebraic form.
(c) Write g in matrix form.
(d) What transformation of $\mathbb{R}^2$ does g represent?
(e) Using the standard basis form, together with linearity, find g^{-1}.
(f) Confirm your answer by finding g^{-1} by: (i) first finding the inverse of the 2×2 matrix from (c); and then (ii) writing this linear map in standard basis form.

L4 · Standard basis form: S and T, and their images and kernels

LA tutor guide Week 7 (R. Carey, 29 Oct 2026): Standard basis form; Lemma 7.9, Remark 7.10

Let e_1, e_2 be the standard basis of $\mathbb{R}^2$.

(a) We are told that $S(e_1) = \frac{1}{2}e_1 + \frac{1}{2}e_2 = S(e_2)$. Is this sufficient to define a linear map $S : \mathbb{R}^2 \to \mathbb{R}^2$? How about $T(e_1) = e_2 = T(e_2)$?

(b) Let

$$T:\ e_1 \mapsto e_2,\ \ e_2 \mapsto e_2, \qquad S:\ e_1 \mapsto \tfrac{1}{2}e_1 + \tfrac{1}{2}e_2,\ \ e_2 \mapsto \tfrac{1}{2}e_1 + \tfrac{1}{2}e_2.$$

Find $\mathrm{Im}(X)$ and $\ker(X)$ where $X = S,\ T,\ S \circ T,\ T \circ S$.

L5 · Image and kernel of two matrices

LA tutor guide Week 7 (R. Carey, 29 Oct 2026): Image and kernel; Fact 7.14, Definition 7.26

Find $\mathrm{Im}(A)$ and $\ker(A)$, i.e. the image and kernel of the map $x \mapsto Ax$, for the matrices

$$A = \begin{pmatrix} 1 & 2 \\ 5 & 4 \end{pmatrix} \qquad \text{and} \qquad A = \begin{pmatrix} 1 & 2 & 6 \\ 5 & 4 & 6 \end{pmatrix}.$$

How is $\ker(A)$ related to the solution set $S(A, 0)$ from LA1a? Explain why the second matrix must have a non-trivial kernel.

L6 · Order of composition: h∘g versus g∘h

LA PS6 Q9(e)

Let $g : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear map with $g(e_1) = e_2$ and $g(e_2) = -e_1$ (Q8), so $g(x) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} x$. Let $h : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear map $h(x) = Ax$ where $A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

(e) By using the matrix forms of the maps, find $h \circ g$ and $g \circ h$ in algebraic form. Are they equal? Describe each one geometrically.

L7 · Bases of the kernel and the image: PS6 Q5(d)

LA PS6 Q5(d)

Compute the kernel and image of the $\mathbb{R}$-linear map $T : \mathbb{R}^3 \to \mathbb{R}^3$, $T(x) = (u_1 \cdot x)v_1 + (u_2 \cdot x)v_2$, where $u_1 = (0, 1, -1)$, $u_2 = (1, 1, 0)$ and $v_1 = (1, 1, 1)$, $v_2 = (1, -1, 0)$. Give a basis of $\operatorname{Im} T$ and $\ker T$.

L8 · Rank–Nullity: the picture, today's examples, and the field

LA tutor guide Week 7 (R. Carey, 29 Oct 2026): Rank–Nullity; Theorem 7.31, Proposition 7.33, Corollary 7.35; PS6 Q4 for Q3(ii)

(a) State the Rank–Nullity Theorem. Picture the domain V as a box (1). Inside it, region (2) is the kernel, which T sends to 0; the rest of the box is mapped onto the image, region (3), inside the codomain W. Explain why "(1) has the same size as (2) and (3) together" is a picture of the theorem.

(b) Verify the theorem for: the maps $S, T, S \circ T, T \circ S$ with $S(e_1) = S(e_2) = \frac{1}{2}e_1 + \frac{1}{2}e_2$ and $T(e_1) = T(e_2) = e_2$; the matrices $\begin{pmatrix} 1 & 2 \\ 5 & 4 \end{pmatrix}$ and $\begin{pmatrix} 1 & 2 & 6 \\ 5 & 4 & 6 \end{pmatrix}$; and the map $T(x) = (u_1 \cdot x)v_1 + (u_2 \cdot x)v_2$ of PS6 Q5(d).

(c) Find a linear map for which (2) = (1). Which maps make (2) as small as possible? What does the size of the codomain W have to do with it?

(d) (PS6 Q4 for Q3(ii).) For $T : \mathbb{C}^4 \to \mathbb{C}^2$, $T(x_1, x_2, x_3, x_4) = (x_1 + ix_2,\ x_3 + ix_4)$, find the rank and nullity first over $\mathbb{C}$ and then over $\mathbb{R}$, and check the theorem both times.

After-class practice

Full solutions to these are on Blackboard with Problem Sheet 6.

R-Q2 · Linear or not? Proof versus counterexample, then the matrix check

LA PS6 Q2(a), (b), (c), (e)

Determine which of the following are $\mathbb{R}$-linear functions.

(a) $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x_1, x_2) = (x_1 - x_2,\ 5x_1)$.

(b) $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x_1, x_2) = (x_1 - x_2^2,\ 5x_1)$.

(c) $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x_1, x_2) = (x_1 - 1,\ 5x_1)$.

(e) $T : \mathbb{R}^2 \to \mathbb{R}$, $T(x_1, x_2) = e^{x_1} - \cos(\pi x_2)$.

For those that are linear, find their matrix form. How could we use this to confirm that those functions were linear?

R-Q2d · A map built from u_1, u_2: its matrix has columns u_1, u_2

LA PS6 Q2(d)

Determine whether the following is an $\mathbb{R}$-linear function: $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(x) = (x \cdot e_1)u_1 + (x \cdot e_2)u_2$, where

$$u_1 = \begin{pmatrix} -1 \\ 3 \end{pmatrix}, \quad u_2 = \begin{pmatrix} 3 \\ 3 \end{pmatrix}, \quad e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

If it is linear, find its matrix form. How could we use this to confirm that the function is linear?

Adapted from Linear Algebra Problem Sheet 6 (University of Bristol, 2026/27) for tutorial use. No solutions are included.